Week 10: Reporting Results, LyX, and LaTex

Samuel Rowe adapted from LyX documentation

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```
bookdown::render_book()
```

Overview

Topics:

1. Introduction to LyX
2. Introduction to LaTex
3. Getting LaTex output from Stata

Chapter 1

Introduction to LyX

We will start off by understanding what LyX is an how we can use it for economic publication quality reports. We will discuss what is LyX and how we can install it onto our machine. Next, we will go through LyX's Tutorial and use the Example document.

1.1 What you see is what you get (WYSIWYG) LaTex word processor

We are able to use LaTex documents without knowing too much LaTex ourselves with LyX. We can insert LaTex code into a LyX document, but with LyX this is not necesary. We will want to understand some of the basic of LaTex, so we can incorporate mathematical equations. We will cover this in the next chapter.

1.2 Install LyX

1.2.1 Windows Installation

Please see: https://wiki.lyx.org/Windows/WindowsSetup

1.2.2 OSX Installation

Please see: https://www.baeldung.com/cs/lyx-installation or https://wiki.lyx.org/Mac/Mac

1.3 LyX Tutorial

After we launch the LyX tutorial, we will want to open a new document. Go to File → New or click on the following:

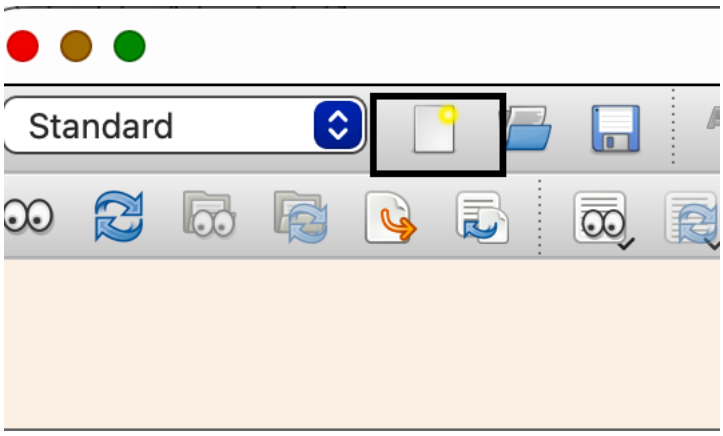


Figure 1.1: New Document

1.3.1 Select Document Class

Go to Document → Settings.

Make sure "Article Standard Class" is selected. This will provide the sections that we need for our document.

Now go back to the main window and look to the upper left to see the Envionment window. It should should the word "Standard".

1.3.2 Sections

After selecting "Standard" in the Environment window, go to "Section" under "Sectioning".

This will create a new section, which will look like this. Next, we can enter our section header, which will be "Introduction" for this example.

Next, press "Enter" and begin to type your introduction paragraph.

1.3.3 Subsections

After your introduction paragraph(s), press enter for a new paragraph. You will notice that the new paragraph is indented. We will type the word "Background"

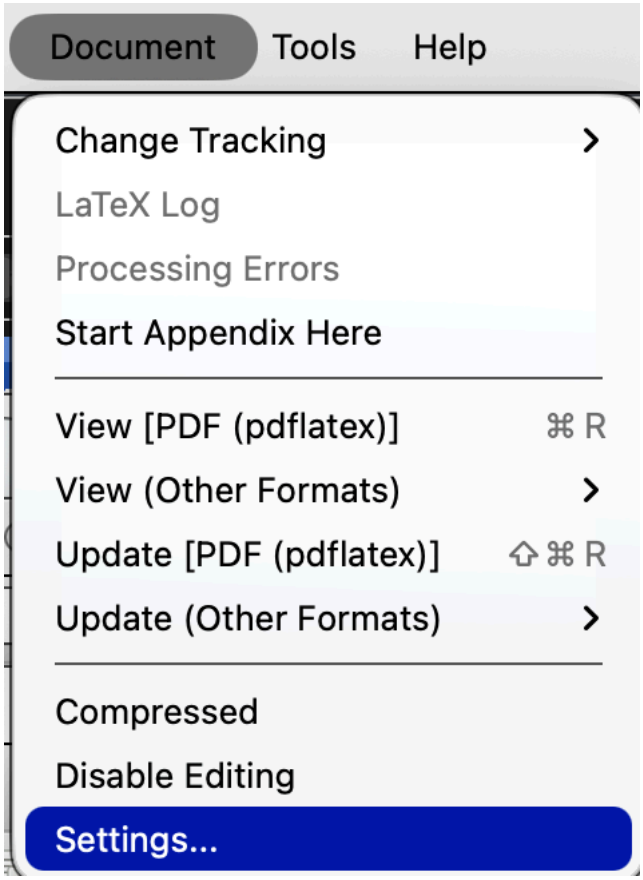


Figure 1.2: Lyx Document Settings

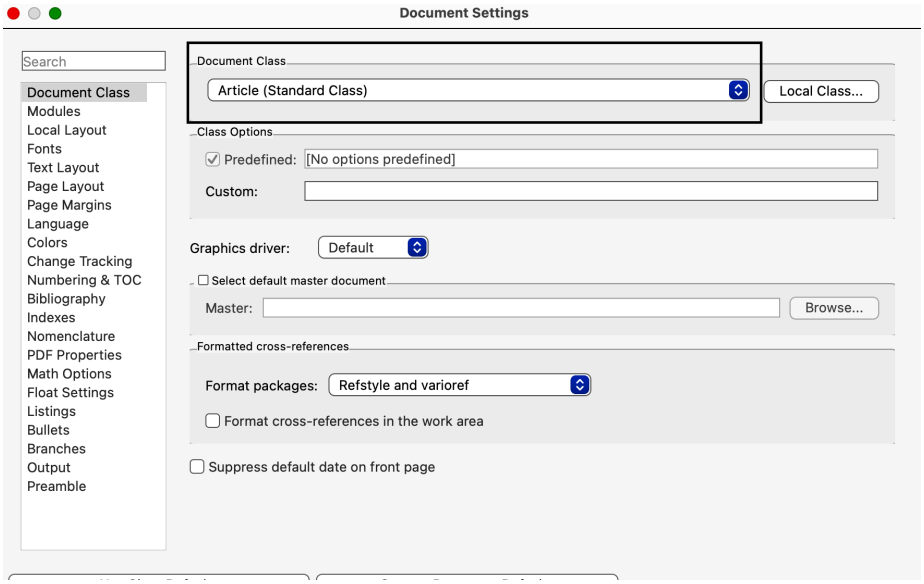


Figure 1.3: Article Standard Class

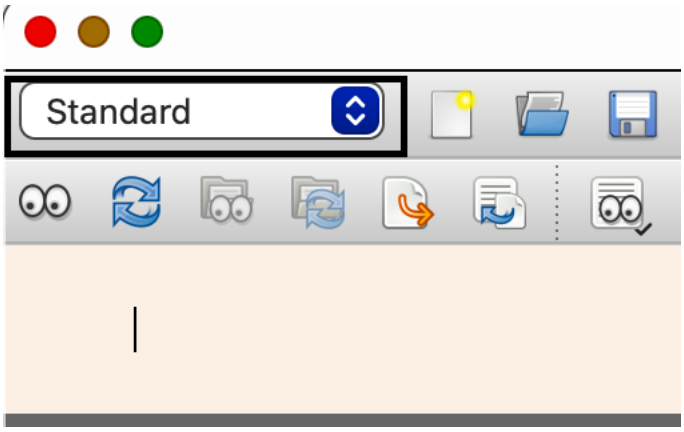


Figure 1.4: LyX Style Window

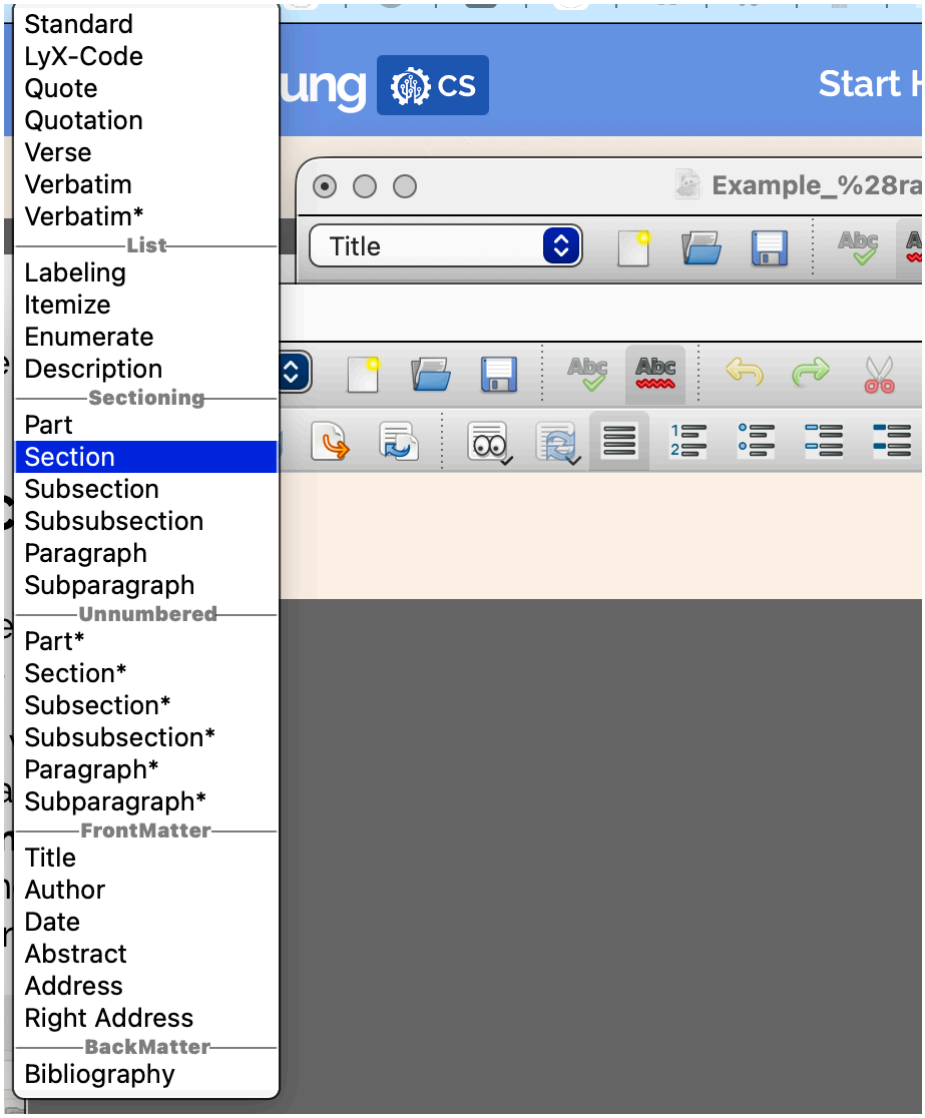


Figure 1.5: LyX Environment Window

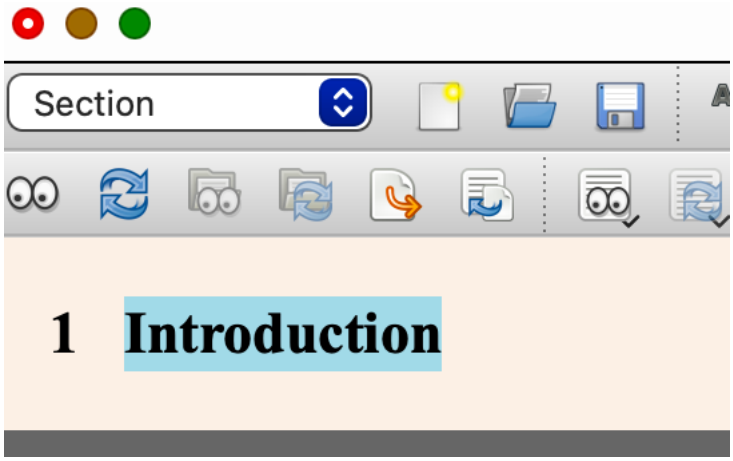


Figure 1.6: LyX Numbered Section 1

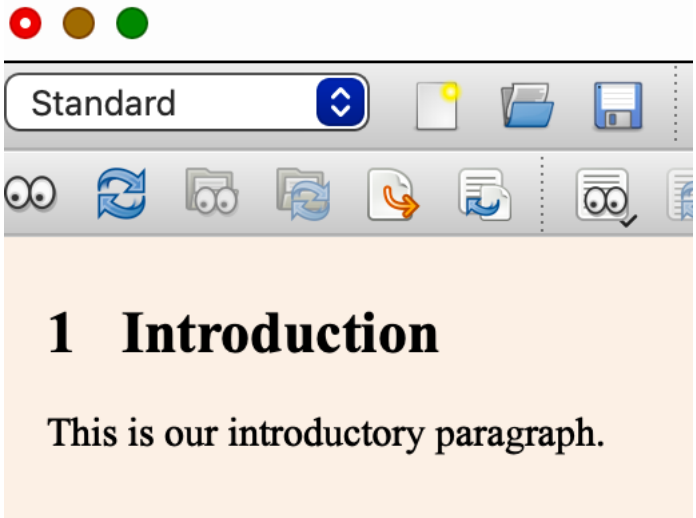


Figure 1.7: Introductory paragraph

for our first subsection. It will look like the following:

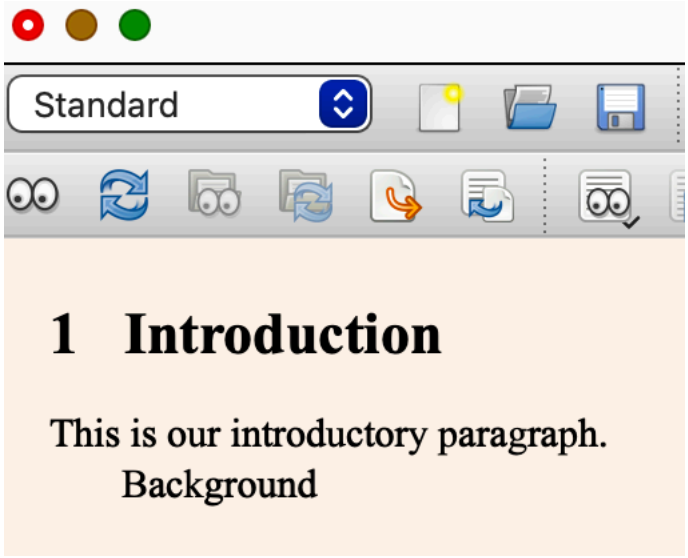


Figure 1.8: New paragraph indented

Now go back to the Environment window and select "Subsection" under "Sectioning".

The result will be that "Background" becomes subsection 1.1.

We can do this several subsections. We will create "Theory" and "Literature Review" subsections.

1.3.4 Preview our work

Now that we have created our first section. Let's preview what our pdf will look like. Click the preview button to see the preview.

1.3.5 Title, Author, and Abstract

Let's add a title to our article. Go to the first line and enter your title of the article. In this example, we will call the article "Econ 643 LyX Example".

Go back to the Environment window and select "Title" under "Front Matter".

Select "Title" and your title will become a title for your article.

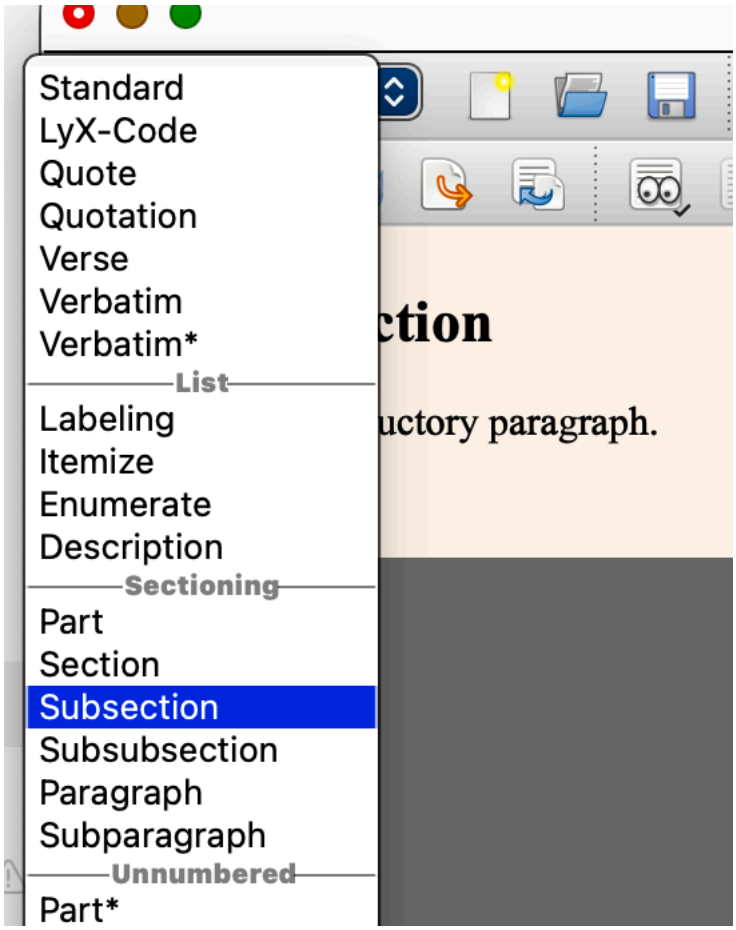


Figure 1.9: New Subsection Selection

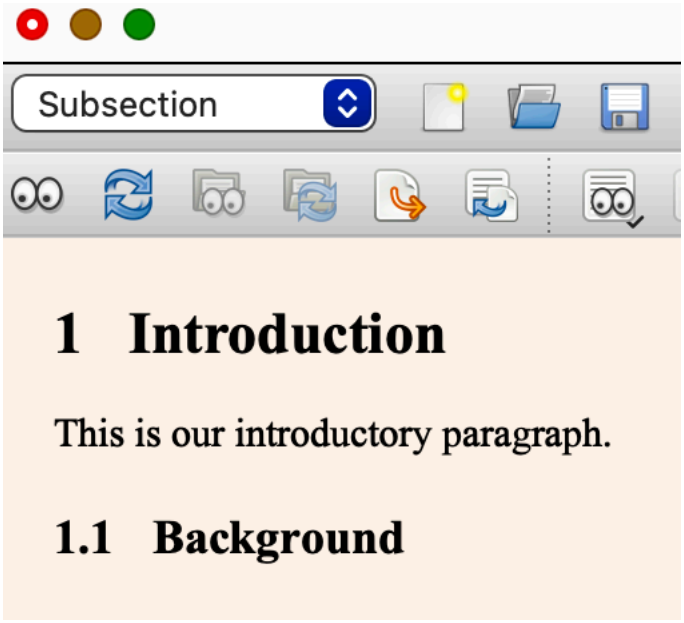


Figure 1.10: New Subsection Added

1.3.6 Add an author

Under your title, add your name.

Next select "Author" under the "Front Matter" section of the Environment window.

1.3.7 Table of Contents

We can add a table of contents (optional). As long as we are using sectioning, a table of contents will automatically generate once we select

1.3.8 Mathematical Equations

We can click the following button to insert a mathematical equation into a LyX document.

1.3.9 Insert LaTex Results

We can export our LaTex code that is produced from Stata with the *estout* commands. When you preview the document, you will notice that there is

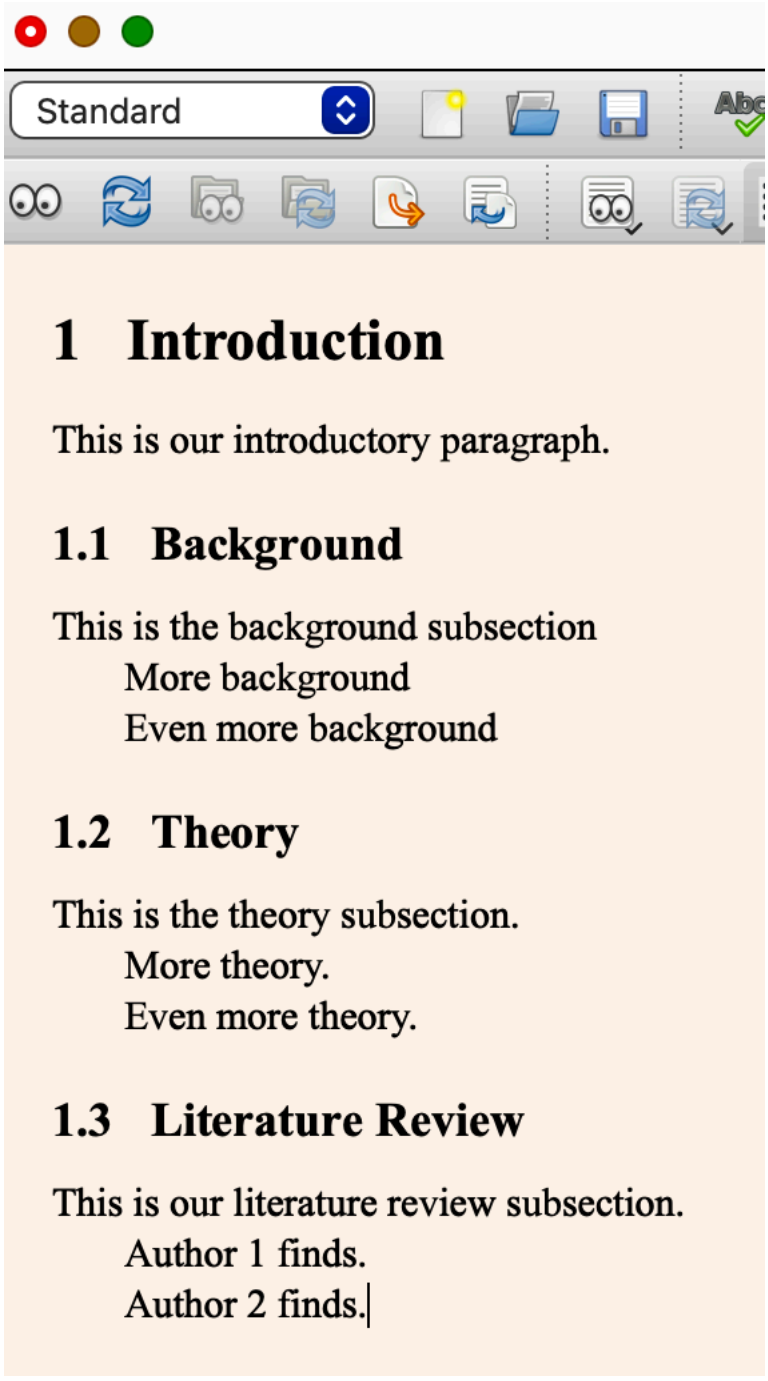


Figure 1.11: Additional subsections

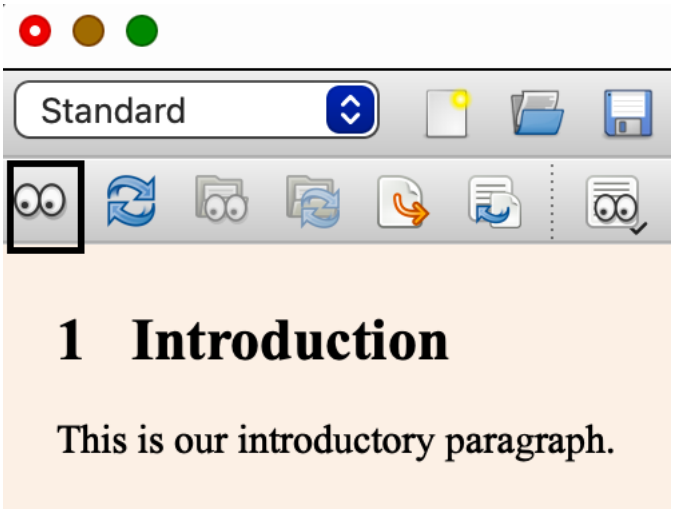


Figure 1.12: Preview your work

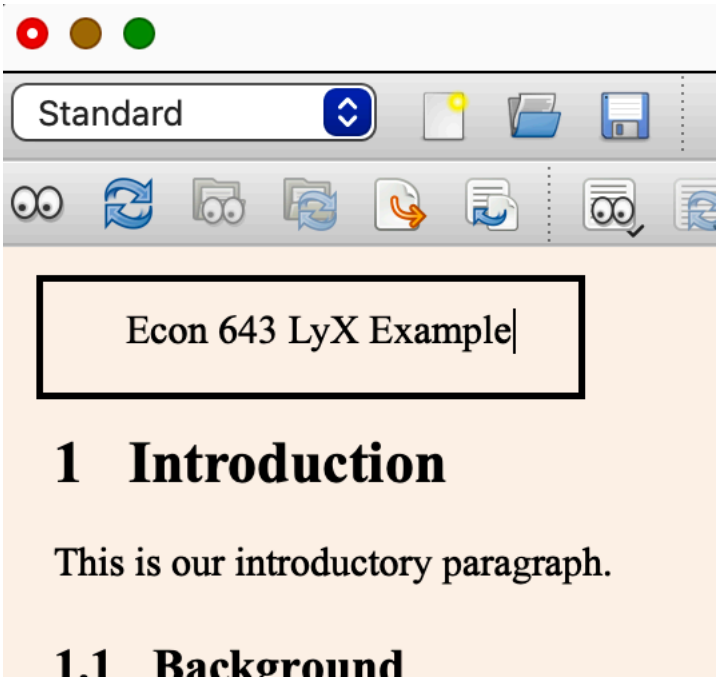


Figure 1.13: Add a Title

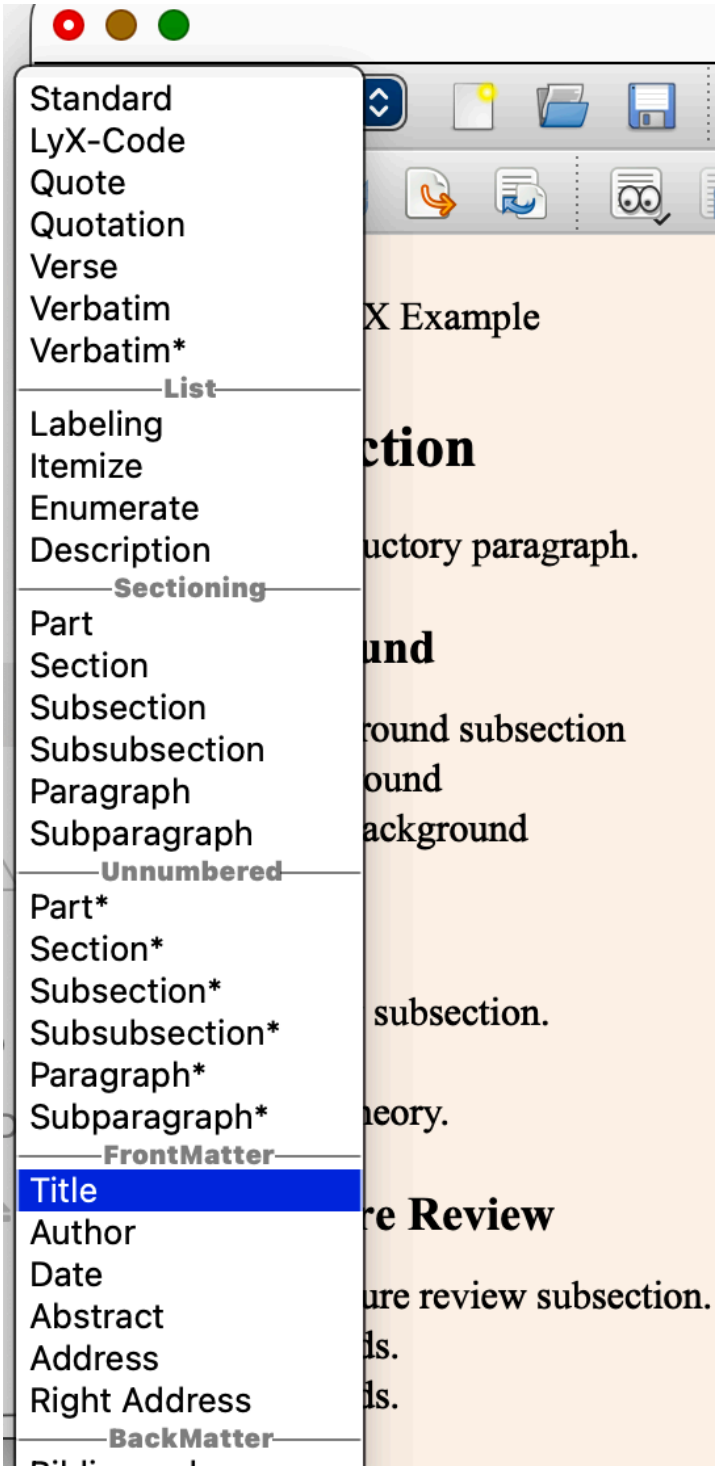


Figure 1.14: Title Option

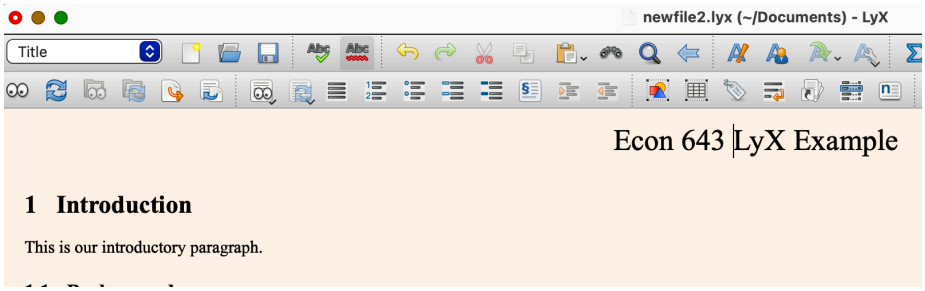


Figure 1.15: New Title

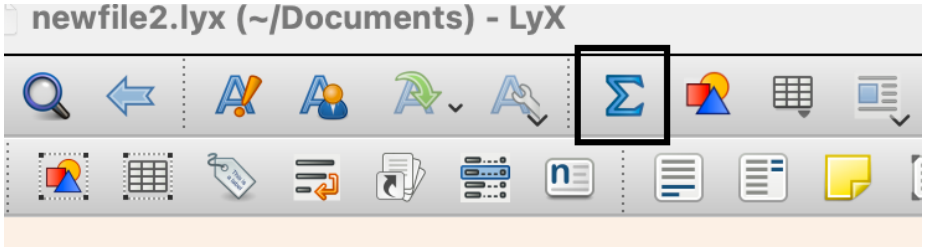


Figure 1.16: Mathematical Equation in LyX

LaTex code that automatically labels the table number.

To insert a place to insert the LaTex output from Stata, click Insert → Tex Code.

Now you will see a new box in your Lyx Document.

Next paste your LaTex code output from Stata into Lyx with right-click and Paste.

We will discuss *estout* in the next chapter.

1.3.10 Insert Figure

Our final tutorial will cover inserting graphs from Stata. We will cover exporting graphs from Stata a bit later.

We need to export our Stata histogram that we created. We will save it as a ".png" file.

Go to Insert → Float → Figure.

Now enter a title for the figure. We will use "Histogram of Automobile Price".

Next, we need to insert the histogram from Stata above the Figure 1: box but below Float Figure. Go to Insert → Graphics.

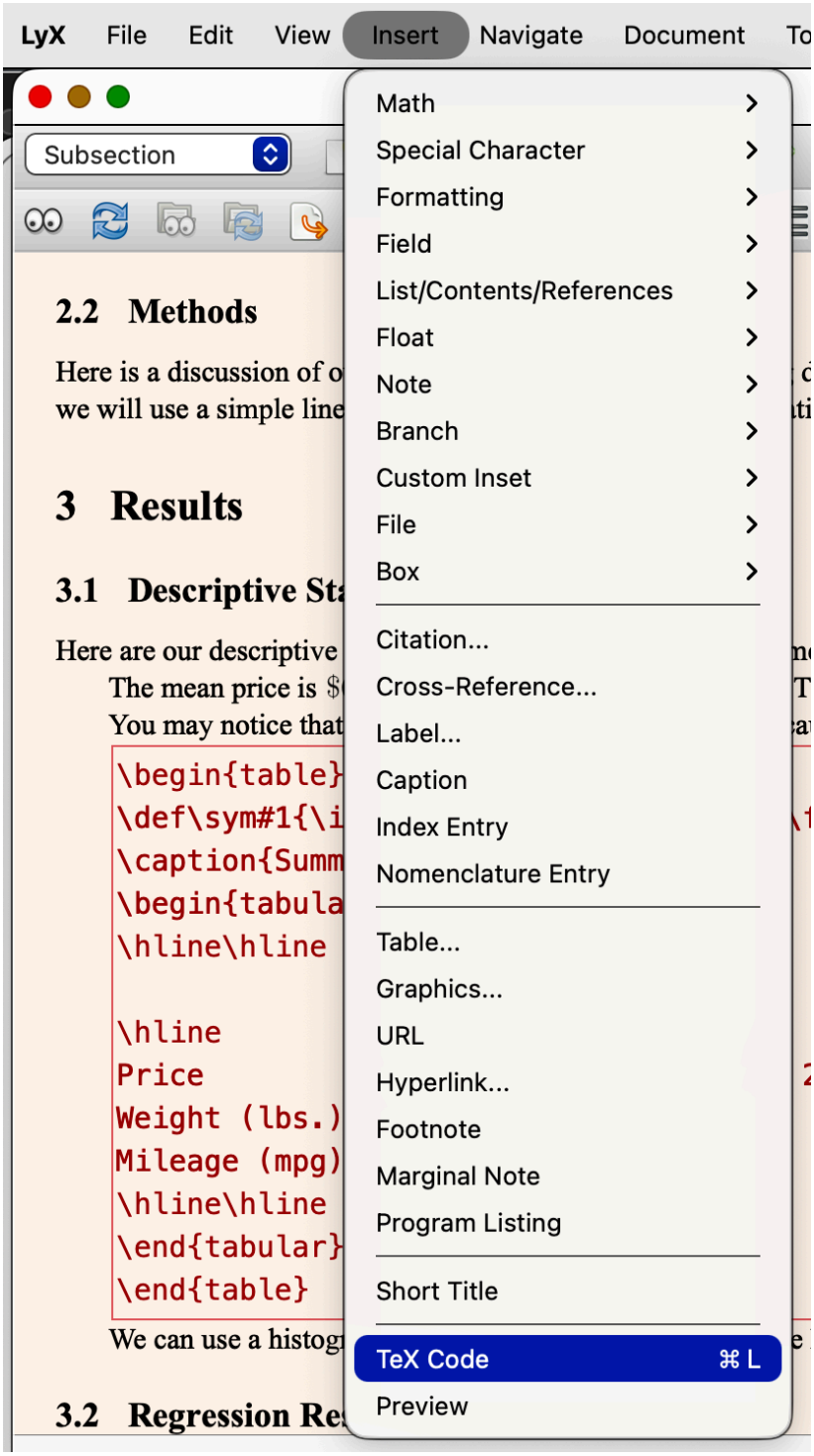


Figure 1.17: Insert Function



Figure 1.18: LaTex insert box

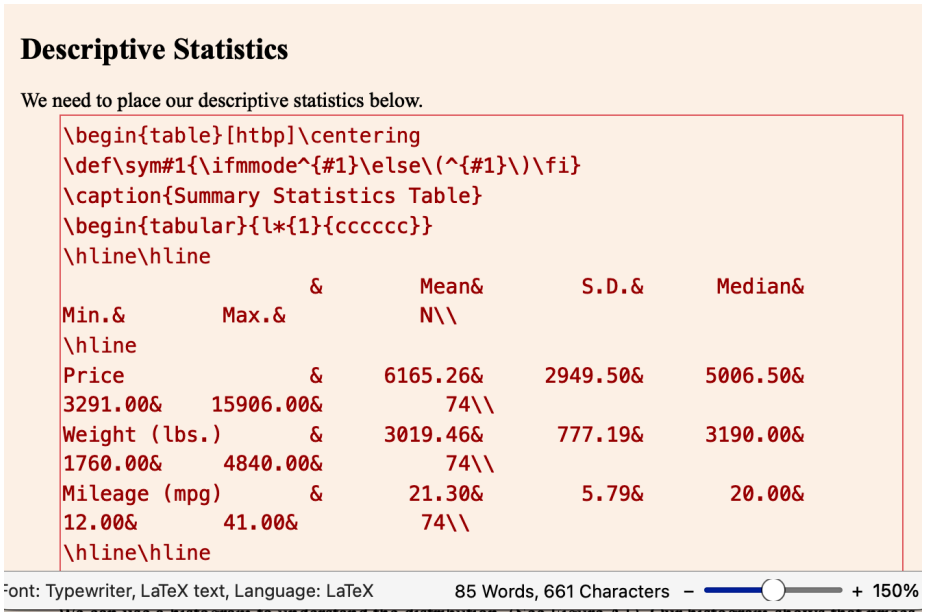


Figure 1.19: LaTex Right-Click Paste

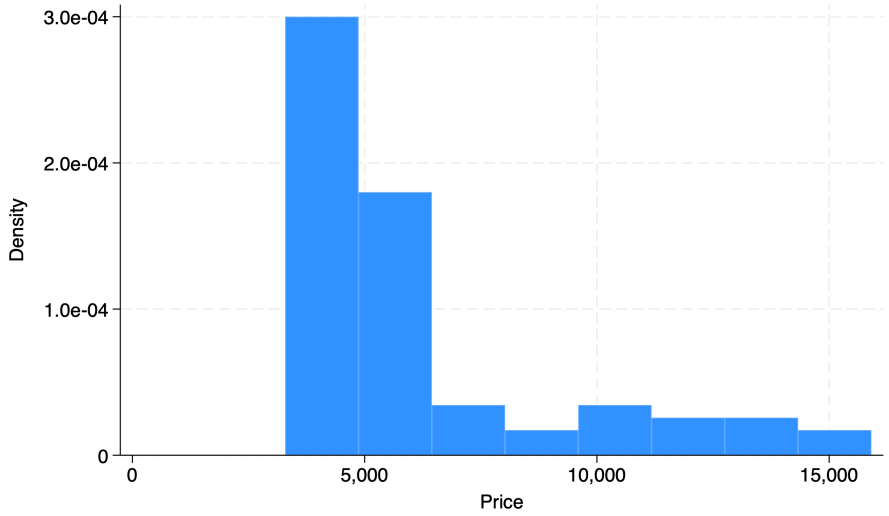


Figure 1.20: Histogram we want to insert

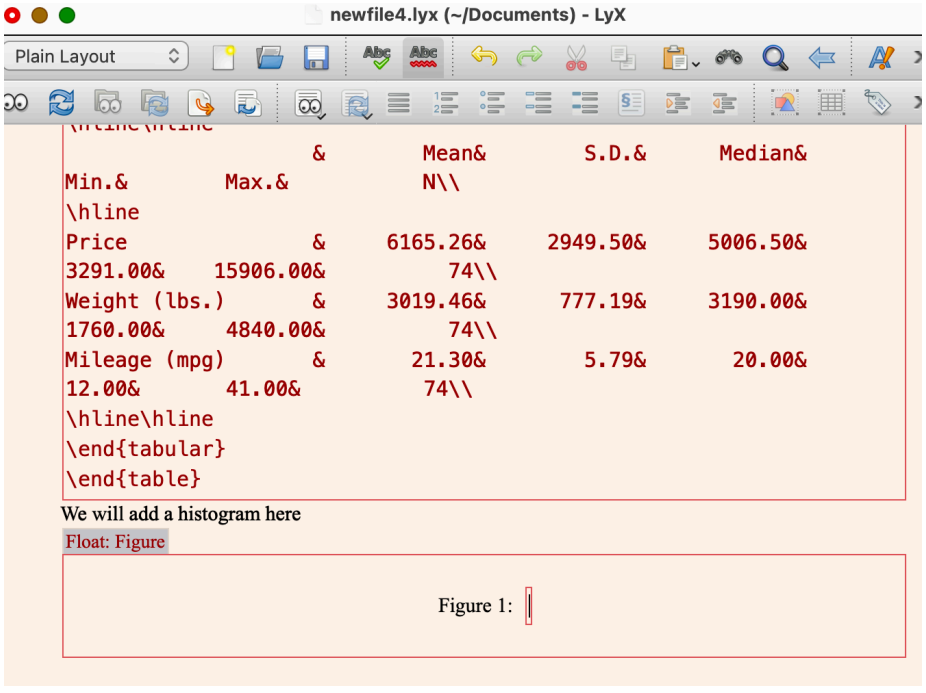


Figure 1.21: Insert Figure

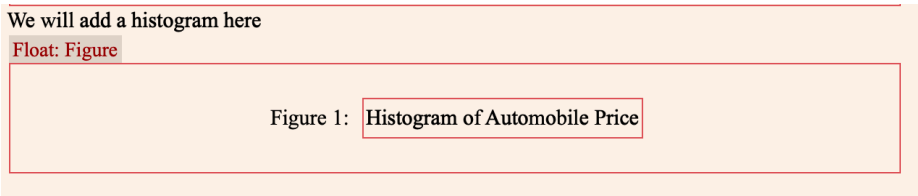


Figure 1.22: Figure Title

In the Graphics window, click browse and find your histogram ".png" file. Next, select "set width" and choose the size of your choice. Next, select "Column Width %" from the drop down menu next to "set width". Click "OK".

Your histogram will be inserted, but it may look funny.

Click on preview to see how well the histogram was inserted. You may need to play around with the settings. I have the size of the Stata graph window makes the graphic file larger.

1.4 Export Your LaTex file into PDF

Go to File → Export → Export As. Select the location of the folder you want to store the file. Next, enter the name of the file and select "PDF (pdflatex) (*.pdf)".

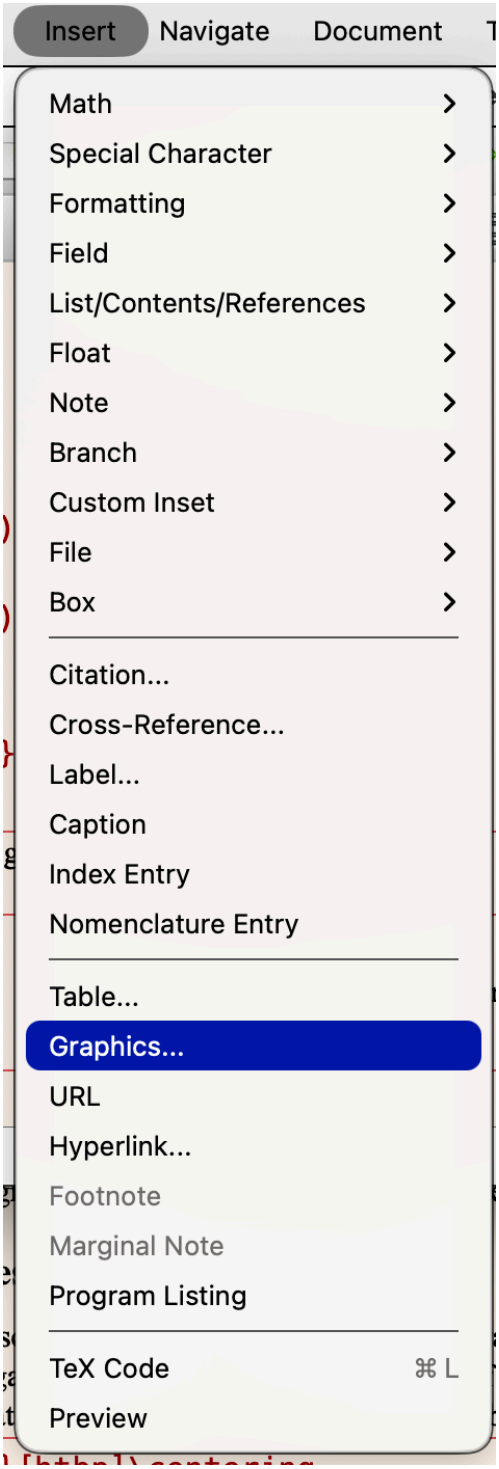


Figure 1.23: Insert Graphics

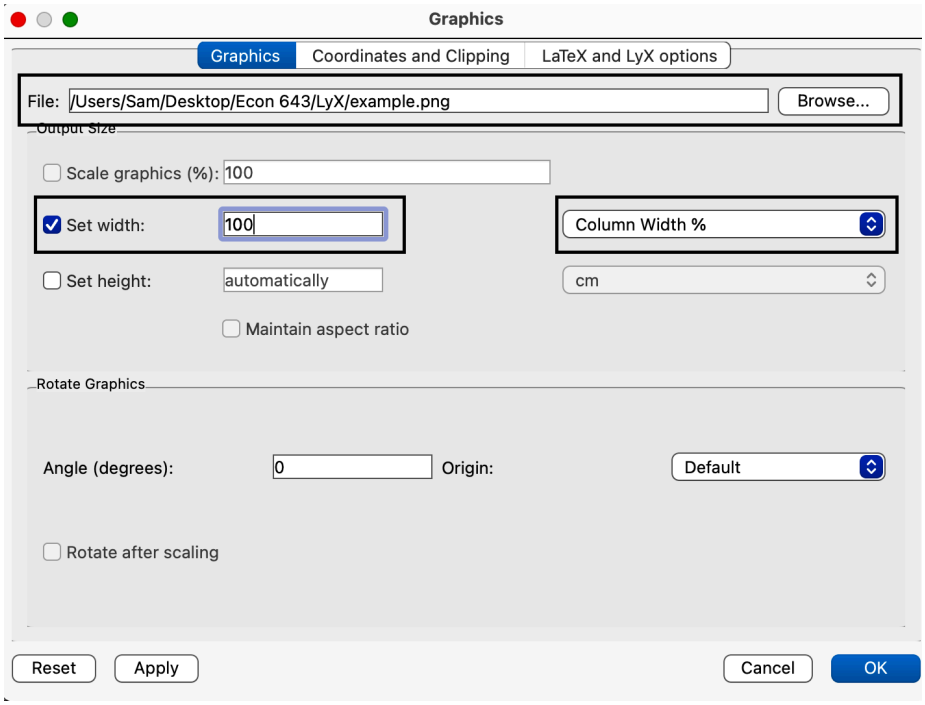


Figure 1.24: Graphics Window

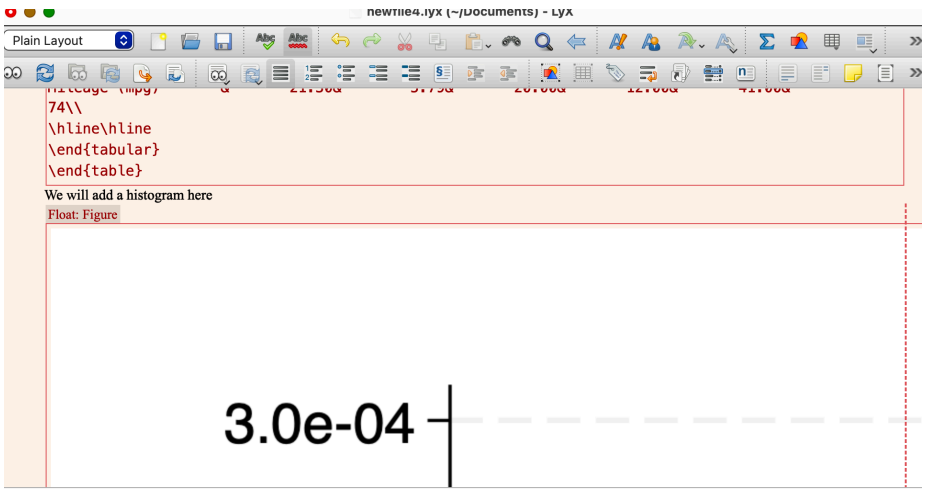


Figure 1.25: Inserted Histogram

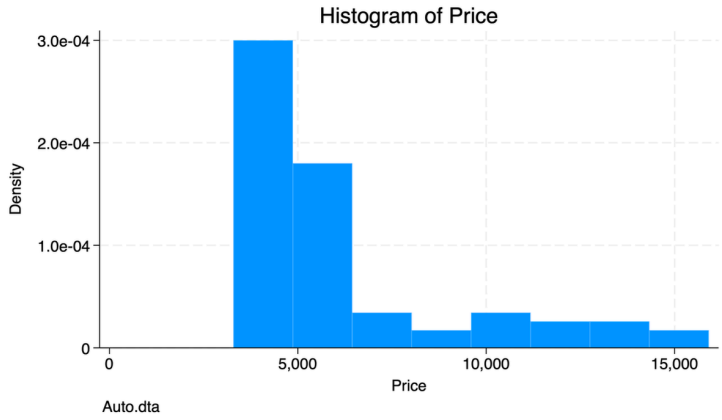


Figure 1: Histogram of Automobile Price

Descriptive Statistics

We need to place our descriptive statistics below.

Figure 1.26: Inserted Histogram PDF

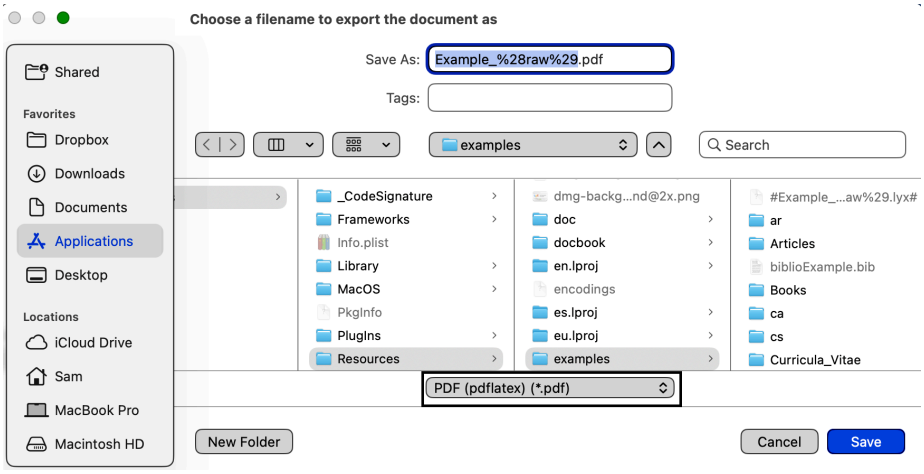


Figure 1.27: Export As in LyX

Chapter 2

Introduction to LaTex

I will provide a crash course on how to use LaTex to create mathematical equations. Kemp, et al. (2016) discuss how LaTeX is the "industry standard" typesetting system for writing mathematics. LaTeX is more flexible than standard word processors, such as MSWord), produces vastly better-looking results.

Learning how to create mathematical equations with LaTex will help you with your homework and projects through the empirical sequence.

With LaTex, we can turn y=bx^2+c into

$$y = \beta x^2 + c$$

If you want a mean, you don't have to type mu or average. Just use LaTex by entering "\ mu" which results in μ. Do you want to expected value of x? You can just use E(x), but it looks better with LaTex with $E(x)$.

2.1 How to use mathematical symbols

We typicall use the "\" to initate a mathemetical symbol. We can use"^" for superscripts and "_" *for subscripts. For example, we can type \ sum^N*{i=1} x_{i} and it will become

$$\sum_{i=1}^{N} x_i$$

You may have noticed that the subscripts had {} brackets. When we want to include an equation in the subscript or subscript, we need to include {} brackets, such as x^{2a+3b}. This will result in x^{2a+3b}

2.2 Overview of mathematical symbols

I think the following is a helpful reference guide for mathematical symbols in LaTex

$\frac{abc}{xyz}$	\frac{abc}{xyz}	$\overline{abc}$	\overline{abc}	$\overrightarrow{abc}$	\overrightarrow{abc}
f'	f'	$\underline{abc}$	\underline{abc}	$\overleftarrow{abc}$	\overleftarrow{abc}
$\sqrt{abc}$	\sqrt{abc}	$\widehat{abc}$	\widehat{abc}	$\overbrace{abc}$	\overbrace{abc}
$\sqrt[n]{abc}$	\sqrt[n]{abc}	$\widetilde{abc}$	\widetilde{abc}	$\underbrace{abc}$	\underbrace{abc}

Source: https://quickref.me/latex.html

2.3 Greek letters

α	\alpha	κ	\kappa	ψ	\psi	$\digamma$	\digamma	Δ	\Delta	Θ	\Theta
β	\beta	λ	\lambda	ρ	\rho	ε	\varepsilon	Γ	\Gamma	Υ	\Upsilon
χ	\chi	μ	\mu	σ	\sigma	$\varkappa$	\varkappa	Λ	\Lambda	Ξ	\Xi
δ	\delta	ν	\nu	τ	\tau	φ	\varphi	Ω	\Omega		
ϵ	\epsilon	o	o	θ	\theta	ϖ	\varpi	Φ	\Phi	$\aleph$	\aleph
η	\eta	ω	\omega	υ	\upsilon	ϱ	\varrho	Π	\Pi	$\beth$	\beth
γ	\gamma	ϕ	\phi	ξ	\xi	ς	\varsigma	Ψ	\Psi	$\daleth$	\daleth
ι	\iota	π	\pi	ζ	\zeta	ϑ	\vartheta	Σ	\Sigma	$\gimel$	\gimel

Source: https://quickref.me/latex.html

2.4 Accents and Braces

Mathematical font

$\mathcal{A}\mathcal{B}\mathcal{C}\mathcal{D}\mathcal{E}\mathcal{F}\mathcal{G}\mathcal{H}\mathcal{I}\mathcal{J}\mathcal{K}\mathcal{L}\mathcal{M}\mathcal{N}\mathcal{O}\mathcal{P}\mathcal{Q}\mathcal{R}\mathcal{S}\mathcal{T}\mathcal{U}\mathcal{V}\mathcal{W}\mathcal{X}\mathcal{Y}\mathcal{Z}$
`\mathcal{` *letter* `}`

Superscript & Subscript

x^y	`x^y`	x^{a+b}	`x^{a+b}`
x_y	`x_y`	x_{a+b}	`x_{a+b}`

Braces

`\overbrace{ ... }^{` *text over brace* `}`
`\underbrace{ ... }_{` *text under brace* `}`

Accents

$\hat{a}$	`\hat{a}`	$\bar{a}$	`\bar{a}`	$\mathring{a}$	`\mathring{a}`
$\check{a}$	`\check{a}`	$\dot{a}$	`\dot{a}`	$\vec{a}$	`\vec{a}`
$\tilde{a}$	`\tilde{a}`	$\ddot{a}$	`\ddot{a}`	$\widehat{AAA}$	`\widehat{AAA}`

Source: https://www.tilburgsciencehub.com/topics/research-skills/templates-dynamic-content/templates/amsmath-latex-cheatsheet/

2.5 Binary Relations and Operators

Source: https://www.tilburgsciencehub.com/topics/research-skills/templates-dynamic-content/templates/amsmath-latex-cheatsheet/

Binary relations

$\neq$	`\ne`	$\leqslant$	`\le`	$\geqslant$	`\ge`
$\equiv$	`\equiv`	$\ll$	`\ll`	$\gg$	`\gg`
$\doteq$	`\doteq`	$\sim$	`\sim`	$\simeq$	`\simeq`
$\subset$	`\subset`	$\supset$	`\supset`	$\approx$	`\approx`
$\subseteq$	`\subseteq`	$\supseteq$	`\supseteq`	$\cong$	`\cong`
$\in$	`\in`	$\ni$	`\ni`	$\propto$	`\propto`
$\mid$	`\mid`	$\parallel$	`\parallel`	$\perp$	`\perp`

It's possible to negate these symbols by prefixing them with `\not` (for example: $\not\equiv$ `\not\equiv`)

Binary operators

$\pm$	`\pm`	$\mp$	`\mp`	$\cdot$	`\cdot`
$\div$	`\div`	$\times$	`\times`	$\setminus$	`\setminus`
$\star$	`\star`	$\cup$	`\cup`	$\cap$	`\cap`
$*$	`\ast`	$\circ$	`\circ`	$\bullet$	`\bullet`
$\oplus$	`\oplus`	$\ominus$	`\ominus`	$\odot$	`\odot`
$\oslash$	`\oslash`	$\otimes$	`\otimes`	$\smallsetminus$	`\smallsetminus`

Figure 2.1: Binary Relations and Operators

Chapter 3

Incorporating LaTex output from Stata

3.1 Table commands

Table command is provides a basis to create customized tables for descriptive statistics. However, we can use the *collect* command after the *table* command to export our results into various documents of our choice.

We have seen the table command before, but let's look at some frequency tables using the *table* command. We use *statistics(frequency)* as the option

```
table (industry) (union), statistic(frequency)
```

	Union member		
	Non-Union	Union	Total
2-digit NAICS			
11-Agriculture	106	7	113
21-Mining	49	7	56
23-Construction	541	78	619
31-33 Mfg Durable	597	45	642
31-33 Mfg Non-durable	290	23	313
42-Wholesale	174	5	179
44-45 Retail	1,016	26	1,042
48-49 Transportation and Warehousing	380	96	476
22-Utilities	82	23	105
51-Information	163	14	177
52-Finance and Insurance	451	6	457

```
  53-Real Estate and Leasing           |        174       3     177
  54-Professional Services             |        741       9     750
  55-56 Management and Mng Services    |        398      15     413
  61-Educational Services              |        648     257     905
  62-Health Care and Social Services   |      1,428     101   1,529
  71-Arts and Entertainment            |        169      14     183
  72-Accommodation and Food Services   |        609      16     625
  814110-Private Households             |         36              36
  81-Other Services                    |        335      11     346
  92-Public Administration             |        440     174     614
  Total                                |      8,827     930   9,757
-------------------------------------------------------------------
```

Next, let's add percentages along with frequencies with *statistic(percent)*.

```
table (industry) (union), statistic(frequency) statistic(percent)
```

```
                                       |         Union member
                                       |  Non-Union   Union    Total
---------------------------------------+----------------------------
2-digit NAICS                          |
  11-Agriculture                       |
    Frequency                          |        106       7      113
    Percent                            |       1.09    0.07     1.16
  21-Mining                            |
    Frequency                          |         49       7       56
    Percent                            |       0.50    0.07     0.57
  23-Construction                      |
    Frequency                          |        541      78      619
    Percent                            |       5.54    0.80     6.34
  31-33 Mfg Durable                    |
    Frequency                          |        597      45      642
    Percent                            |       6.12    0.46     6.58
  31-33 Mfg Non-durable                |
    Frequency                          |        290      23      313
    Percent                            |       2.97    0.24     3.21
  42-Wholesale                         |
    Frequency                          |        174       5      179
    Percent                            |       1.78    0.05     1.83
  44-45 Retail                         |
    Frequency                          |      1,016      26    1,042
    Percent                            |      10.41    0.27    10.68
  48-49 Transportation and Warehousing |
    Frequency                          |        380      96      476
    Percent                            |       3.89    0.98     4.88
```

22-Utilities			
Frequency	82	23	105
Percent	0.84	0.24	1.08
51-Information			
Frequency	163	14	177
Percent	1.67	0.14	1.81
52-Finance and Insurance			
Frequency	451	6	457
Percent	4.62	0.06	4.68
53-Real Estate and Leasing			
Frequency	174	3	177
Percent	1.78	0.03	1.81
54-Professional Services			
Frequency	741	9	750
Percent	7.59	0.09	7.69
55-56 Management and Mng Services			
Frequency	398	15	413
Percent	4.08	0.15	4.23
61-Educational Services			
Frequency	648	257	905
Percent	6.64	2.63	9.28
62-Health Care and Social Services			
Frequency	1,428	101	1,529
Percent	14.64	1.04	15.67
71-Arts and Entertainment			
Frequency	169	14	183
Percent	1.73	0.14	1.88
72-Accommodation and Food Services			
Frequency	609	16	625
Percent	6.24	0.16	6.41
814110-Private Households			
Frequency	36		36
Percent	0.37		0.37
81-Other Services			
Frequency	335	11	346
Percent	3.43	0.11	3.55
92-Public Administration			
Frequency	440	174	614
Percent	4.51	1.78	6.29
Total			
Frequency	8,827	930	9,757
Percent	90.47	9.53	100.00

Let's add mean, median, and standard deviation with statistic(mean) statistic(median) and statistic(sd) as options.

```
table (industry) (union) () if prerelg==1, statistic(frequency) statistic(mean earnings
```

	Union member		
	Non-Union	Union	Total
2-digit NAICS			
11-Agriculture			
Frequency	106	7	113
Mean	1072.304	927.4286	1063.329
Standard deviation	1080.452	434.2207	1051.545
Median	820	976	840
21-Mining			
Frequency	49	7	56
Mean	2369.89	1546.857	2267.011
Standard deviation	2303.516	367.9363	2172.799
Median	1630	1550	1550
23-Construction			
Frequency	541	78	619
Mean	1463.722	1991.516	1530.229
Standard deviation	1346.098	1197.949	1338.961
Median	1154	1820	1200
31-33 Mfg Durable			
Frequency	597	45	642
Mean	1966.478	1751.644	1951.419
Standard deviation	1991.125	880.8464	1934.561
Median	1400	1560	1400
31-33 Mfg Non-durable			
Frequency	290	23	313
Mean	1643.923	1923.053	1664.434
Standard deviation	1724.473	1999.506	1744.081
Median	1200	1500	1220
42-Wholesale			
Frequency	174	5	179
Mean	1711.341	1356.8	1701.437
Standard deviation	1862.348	447.3781	1838.164
Median	1245	1440	1250
44-45 Retail			
Frequency	1,016	26	1,042
Mean	1157.987	916.2308	1151.955
Standard deviation	1559.636	615.6181	1543.449
Median	800	728	800
48-49 Transportation and Warehousing			
Frequency	380	96	476
Mean	1387.652	1941.617	1499.376
Standard deviation	1594.414	2328.189	1778.195

Median	1000	1265	1058
22-Utilities			
Frequency	82	23	105
Mean	2140.056	1993.304	2107.911
Standard deviation	1917.11	914.6595	1744.475
Median	1540	1650	1550
51-Information			
Frequency	163	14	177
Mean	2428.985	2115.015	2404.151
Standard deviation	2542.707	2581.204	2539.766
Median	1600	1405	1540
52-Finance and Insurance			
Frequency	451	6	457
Mean	2422.176	1864.667	2414.857
Standard deviation	2561.741	966.0283	2547.633
Median	1540	1590	1540
53-Real Estate and Leasing			
Frequency	174	3	177
Mean	1629.756	1680	1630.607
Standard deviation	2012.841	1873.286	2005.589
Median	1134	700	1134
54-Professional Services			
Frequency	741	9	750
Mean	2508.094	2145.556	2503.744
Standard deviation	2535.098	686.4967	2521.129
Median	1770	1920	1770
55-56 Management and Mng Services			
Frequency	398	15	413
Mean	1164.696	1128.933	1163.397
Standard deviation	1213.527	607.0499	1196.494
Median	880	1120	880
61-Educational Services			
Frequency	648	257	905
Mean	1424.192	1599.636	1474.014
Standard deviation	1441.982	1067.012	1347.908
Median	1116	1380	1210
62-Health Care and Social Services			
Frequency	1,428	101	1,529
Mean	1438.731	1984.624	1474.79
Standard deviation	1663.149	2149.192	1704.088
Median	1000	1480	1020
71-Arts and Entertainment			
Frequency	169	14	183
Mean	935.1741	1874.444	1007.031
Standard deviation	1232.381	2654.074	1402.768
Median	744	1087	760

```
  72-Accommodation and Food Services   |
    Frequency                          |        609         16        625
    Mean                               |   748.7192      977.5    754.576
    Standard deviation                 |   620.1688    511.745    618.345
    Median                             |        586        840        600
  814110-Private Households             |
    Frequency                          |         36                   36
    Mean                               |   597.8889              597.8889
    Standard deviation                 |    452.714               452.714
    Median                             |        451                   451
  81-Other Services                    |
    Frequency                          |        335         11        346
    Mean                               |   1360.505   2656.565   1401.709
    Standard deviation                 |    1487.65   2737.656   1552.944
    Median                             |       1000       1580       1000
  92-Public Administration             |
    Frequency                          |        440        174        614
    Mean                               |   1893.953   1773.373   1859.782
    Standard deviation                 |   1873.156   1512.013   1777.899
    Median                             |       1400       1535       1440
  Total                                |
    Frequency                          |      8,827        930      9,757
    Mean                               |   1568.534   1754.695   1586.278
    Standard deviation                 |   1817.987   1565.957   1796.251
    Median                             |       1066       1440       1120
-------------------------------------------------------------------------
```

3.1.1 Nesting within rows

We can nest the two categorical variables within rows instead of one categorical being rows and the other categorical being columns.

```
table (industry union) () () if prerelg==1, statistic(frequency) statistic(mean earning
```

```
                                     |  Frequency       Mean   Standard deviation   M
-------------------------------------+-----------------------------------------------
2-digit NAICS                        |
  11-Agriculture                     |
    Union member                     |
      Non-Union                      |        106   1072.304             1080.452
      Union                          |          7   927.4286             434.2207
  21-Mining                          |
    Union member                     |
      Non-Union                      |         49    2369.89             2303.516
```

Union	7	1546.857	367.9363	1550
23-Construction				
Union member				
Non-Union	541	1463.722	1346.098	1154
Union	78	1991.516	1197.949	1820
31-33 Mfg Durable				
Union member				
Non-Union	597	1966.478	1991.125	1400
Union	45	1751.644	880.8464	1560
31-33 Mfg Non-durable				
Union member				
Non-Union	290	1643.923	1724.473	1200
Union	23	1923.053	1999.506	1500
42-Wholesale				
Union member				
Non-Union	174	1711.341	1862.348	1245
Union	5	1356.8	447.3781	1440
44-45 Retail				
Union member				
Non-Union	1,016	1157.987	1559.636	800
Union	26	916.2308	615.6181	728
48-49 Transportation and Warehousing				
Union member				
Non-Union	380	1387.652	1594.414	1000
Union	96	1941.617	2328.189	1265
22-Utilities				
Union member				
Non-Union	82	2140.056	1917.11	1540
Union	23	1993.304	914.6595	1650
51-Information				
Union member				
Non-Union	163	2428.985	2542.707	1600
Union	14	2115.015	2581.204	1405
52-Finance and Insurance				
Union member				
Non-Union	451	2422.176	2561.741	1540
Union	6	1864.667	966.0283	1590
53-Real Estate and Leasing				
Union member				
Non-Union	174	1629.756	2012.841	1134
Union	3	1680	1873.286	700
54-Professional Services				
Union member				
Non-Union	741	2508.094	2535.098	1770
Union	9	2145.556	686.4967	1920
55-56 Management and Mng Services				

```
    Union member                        |
      Non-Union                         |        398   1164.696             1213.527
      Union                             |         15   1128.933             607.0499
  61-Educational Services               |
    Union member                        |
      Non-Union                         |        648   1424.192             1441.982
      Union                             |        257   1599.636             1067.012
  62-Health Care and Social Services    |
    Union member                        |
      Non-Union                         |      1,428   1438.731             1663.149
      Union                             |        101   1984.624             2149.192
  71-Arts and Entertainment             |
    Union member                        |
      Non-Union                         |        169   935.1741             1232.381
      Union                             |         14   1874.444             2654.074
  72-Accommodation and Food Services    |
    Union member                        |
      Non-Union                         |        609   748.7192             620.1688
      Union                             |         16      977.5              511.745
  814110-Private Households              |
    Union member                        |
      Non-Union                         |         36   597.8889              452.714
  81-Other Services                     |
    Union member                        |
      Non-Union                         |        335   1360.505              1487.65
      Union                             |         11   2656.565             2737.656
  92-Public Administration              |
    Union member                        |
      Non-Union                         |        440   1893.953             1873.156
      Union                             |        174   1773.373             1512.013
------------------------------------------------------------------------------------
```

We can flip the nesting order by reversing the variable order within the row parantheses.

```
table (union industry) () () if prerelg==1, statistic(frequency) statistic(mean earning
```

```
                                        |  Frequency       Mean   Standard deviatio
----------------------------------------+-------------------------------------------
Union member                            |
  Non-Union                             |
    2-digit NAICS                       |
      11-Agriculture                    |        106   1072.304             1080.45
      21-Mining                         |         49    2369.89             2303.51
      23-Construction                   |        541   1463.722             1346.09
```

```
      31-33 Mfg Durable                   |        597   1966.478            1991.125     1400
      31-33 Mfg Non-durable               |        290   1643.923            1724.473     1200
      42-Wholesale                        |        174   1711.341            1862.348     1245
      44-45 Retail                        |      1,016   1157.987            1559.636      800
      48-49 Transportation and Warehousing |        380   1387.652            1594.414     1000
      22-Utilities                        |         82   2140.056             1917.11     1540
      51-Information                      |        163   2428.985            2542.707     1600
      52-Finance and Insurance            |        451   2422.176            2561.741     1540
      53-Real Estate and Leasing          |        174   1629.756            2012.841     1134
      54-Professional Services            |        741   2508.094            2535.098     1770
      55-56 Management and Mng Services   |        398   1164.696            1213.527      880
      61-Educational Services             |        648   1424.192            1441.982     1116
      62-Health Care and Social Services  |      1,428   1438.731            1663.149     1000
      71-Arts and Entertainment           |        169   935.1741            1232.381      744
      72-Accommodation and Food Services  |        609   748.7192            620.1688      586
      814110-Private Households           |         36   597.8889             452.714      451
      81-Other Services                   |        335   1360.505             1487.65     1000
      92-Public Administration            |        440   1893.953            1873.156     1400
  Union                                   |
    2-digit NAICS                         |
      11-Agriculture                      |          7   927.4286            434.2207      976
      21-Mining                           |          7   1546.857            367.9363     1550
      23-Construction                     |         78   1991.516            1197.949     1820
      31-33 Mfg Durable                   |         45   1751.644            880.8464     1560
      31-33 Mfg Non-durable               |         23   1923.053            1999.506     1500
      42-Wholesale                        |          5    1356.8            447.3781     1440
      44-45 Retail                        |         26   916.2308            615.6181      728
      48-49 Transportation and Warehousing |         96   1941.617            2328.189     1265
      22-Utilities                        |         23   1993.304            914.6595     1650
      51-Information                      |         14   2115.015            2581.204     1405
      52-Finance and Insurance            |          6   1864.667            966.0283     1590
      53-Real Estate and Leasing          |          3      1680            1873.286      700
      54-Professional Services            |          9   2145.556            686.4967     1920
      55-56 Management and Mng Services   |         15   1128.933            607.0499     1120
      61-Educational Services             |        257   1599.636            1067.012     1380
      62-Health Care and Social Services  |        101   1984.624            2149.192     1480
      71-Arts and Entertainment           |         14   1874.444            2654.074     1087
      72-Accommodation and Food Services  |         16     977.5             511.745      840
      81-Other Services                   |         11   2656.565            2737.656     1580
      92-Public Administration            |        174   1773.373            1512.013     1535
-------------------------------------------------------------------------------------------
```

3.1.2 Nesting within columns

We can nest the two categorical variables within columns as well.

```
table () (union sex) () if prerelg==1, statistic(frequency) statistic(mean earnings) st
```

```
                  |                Union member
                  |       Non-Union               Union
                  |    Individual's Sex      Individual's Sex
                  |      Male     Female       Male     Female
------------------+-----------------------------------------
Frequency         |     4,509      4,318        516        414
Mean              |  1800.009   1326.819    1968.19   1488.601
Standard deviation |  2021.755   1540.911   1851.626   1054.904
Median            |      1220        962       1540       1300
-------------------------------------------------------------
```

We can reverse the order of the nesting within columns as well.

```
table () (sex union) () if prerelg==1, statistic(frequency) statistic(percent) statisti
```

```
                            |               Individual's Sex
                            |          Male                  Female
                            |      Union member           Union member
                            |  Non-Union      Union   Non-Union      Union
----------------------------+-----------------------------------------------
Frequency                   |      4,509        516       4,318        414
Percent                     |      46.21       5.29       44.26       4.24
Mean                        |
  Weekly earnings in dollars |   1800.009    1968.19    1326.819   1488.601
Standard deviation          |
  Weekly earnings in dollars |   2021.755   1851.626    1540.911   1054.904
Median                      |
  Weekly earnings in dollars |       1220       1540         962       1300
-------------------------------------------------------------------------------
```

3.1.3 Customize the format

We can customize the formats with *sformat* to add string characters and *nformat* to format the numerics.

```
table () (sex union) () if prerelg==1, statistic(frequency) statistic(percent) statisti
```

```
                            |               Individual's Sex
                            |          Male                  Female
                            |      Union member           Union member
```

```
                            |  Non-Union      Union   Non-Union      Union
----------------------------+-----------------------------------------------
Frequency                   |      4,509        516       4,318        414
Percent                     |     46.21%      5.29%      44.26%      4.24%
Mean                        |
  Weekly earnings in dollars |   $1800.01   $1968.19    $1326.82   $1488.60
Standard deviation          |
  Weekly earnings in dollars |   $2021.76   $1851.63    $1540.91   $1054.90
Median                      |
  Weekly earnings in dollars |   $1220.00   $1540.00     $962.00   $1300.00
-------------------------------------------------------------------------------
```

3.2 Collect commands

Stata has a collection of *collect* commands to help you format and export your descriptive statistics after the *table* command.

We can preview our table comments with the *collect preview* command.

```
collect preview
```

```
                            |                Individual's Sex
                            |          Male                   Female
                            |      Union member           Union member
                            |  Non-Union      Union   Non-Union      Union
----------------------------+-----------------------------------------------
Frequency                   |      4,509        516       4,318        414
Percent                     |     46.21%      5.29%      44.26%      4.24%
Mean                        |
  Weekly earnings in dollars |   $1800.01   $1968.19    $1326.82   $1488.60
Standard deviation          |
  Weekly earnings in dollars |   $2021.76   $1851.63    $1540.91   $1054.90
Median                      |
  Weekly earnings in dollars |   $1220.00   $1540.00     $962.00   $1300.00
-------------------------------------------------------------------------------
```

We can change our labels if we want with the *modify* option. We can check the change with the *collect label list* command.

```
collect label dim sex "Sex of Individual", modify
collect label list sex
collect preview
```

```
  Collection: Table
   Dimension: sex
       Label: Sex of Individual
Level labels:
          .m  Total
           0  Male
           1  Female
```

```
------------------------------------------------------------------------------
                             |               Sex of Individual
                             |          Male                   Female
                             |      Union member           Union member
                             |  Non-Union      Union   Non-Union      Union
-----------------------------+------------------------------------------------
Frequency                    |      4,509        516       4,318        414
Percent                      |     46.21%      5.29%      44.26%      4.24%
Mean                         |
  Weekly earnings in dollars |   $1800.01   $1968.19    $1326.82   $1488.60
Standard deviation           |
  Weekly earnings in dollars |   $2021.76   $1851.63    $1540.91   $1054.90
Median                       |
  Weekly earnings in dollars |   $1220.00   $1540.00     $962.00   $1300.00
------------------------------------------------------------------------------
```

Let's modify the other variable, too.

```
collect label dim union "Union Status", modify
collect label list union
collect preview
```

```
  Collection: Table
   Dimension: union
       Label: Union Status
Level labels:
          .m  Total
           0  Non-Union
           1  Union
```

```
------------------------------------------------------------------------------
                             |               Sex of Individual
                             |          Male                   Female
                             |      Union Status           Union Status
                             |  Non-Union      Union   Non-Union      Union
```

```
----------------------------+-------------------------------------------
Frequency                   |     4,509        516       4,318        414
Percent                     |    46.21%      5.29%      44.26%      4.24%
Mean                        |
  Weekly earnings in dollars |  $1800.01   $1968.19    $1326.82   $1488.60
Standard deviation          |
  Weekly earnings in dollars |  $2021.76   $1851.63    $1540.91   $1054.90
Median                      |
  Weekly earnings in dollars |  $1220.00   $1540.00     $962.00   $1300.00
-------------------------------------------------------------------------
```

We can use the *collect dims* command to get the collect dimensions.

```
collect dims
collect label list result, all
```

```
Collection dimensions
Collection: Table
------------------------------------------
                     Dimension   No. levels
------------------------------------------
Layout, style, header, label
                        across   2
                        cmdset   1
                       colname   1
                       command   1
                        result   5
                           sex   2
                       statcmd   5
                         union   2
                           var   2

Style only
                  border_block   4
                     cell_type   4
          header_cell_contents   3
------------------------------------------


  Collection: Table
   Dimension: result
       Label: Result
Level labels:
   frequency  Frequency
        mean  Mean
```

```
      median  Median
     percent  Percent
          sd  Standard deviation
```

We can export our descriptive statistics table into various formats. Let's export to LaTex.

```
collect export example.tex, replace
```

Let's use *collect export* into Markdown.

```
collect export example.md, replace
```

Let's use *collect export* into Excel. We have additional commands with *putexcel* to generate a report template and then use *collect export* to export the descriptive statistics table.

```
collect export example.xlsx, replace
```

Let's use *collect export* into MS Word. We have additional commands with *putdocx* to generate a report template and then use *collect export* to export the descriptive statistics table.

```
collect export example.docx, replace
```

There is also a suite of commands with *putpdf* to export tables into a pdf file

Chapter 4

References

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